# Braillix - Slice It Yourself (Anycubic Kobra Neo)

> A complete guide to producing your own G-code on your PC, so the lab just loads the file. Written 2026-07-30 for:
> **Anycubic Kobra Neo - 0.4mm nozzle - PETG - lab accepts G-code.**

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## Why your first print failed - the three real causes

| What you saw | Actual cause |
| --- | --- |
| Supports welded in, couldn't pluck them out | Supports were switched **on** with PLA-tuned settings. PETG bonds across a gap that PLA snaps at. **These parts need no supports at all.** |
| Screw columns snapped | Not enough **walls**. On a dia 7.8mm column, walls are almost the whole part - infill barely matters. |
| Filament strings between travel moves | **Stringing.** Wet PETG + too-hot nozzle + wrong retraction. The Kobra Neo is **direct drive**, so most online retraction advice (5-6mm, meant for Bowden) is wrong and makes it worse. |

None of these are your fault, and only one is the teacher's - they're the default settings.

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# PART 1 - Software

## Install OrcaSlicer (recommended)

**Download:** <https://github.com/SoftFever/OrcaSlicer/releases> -> the `.exe` for Windows.

**Why OrcaSlicer over Cura, for your situation:**

- Anycubic Kobra profiles are **built in** - far fewer ways to get the machine setup wrong
- Has **built-in calibration tests** (retraction, temperature) that directly fix your stringing
- Per-object settings are much easier - which you need for the screw columns

> Cura also works and your teacher may know it better. Cura equivalents for every setting are in **Appendix A** at the end.

## First-run setup

1. Launch OrcaSlicer -> the setup wizard appears
2. **Printer** -> search `Anycubic` -> pick **Kobra Neo** (or **Kobra 2 Neo** - check the machine's

boot screen for which one). If it is genuinely absent, use **Appendix B** to add it manually.

1. **Filament** -> tick **Generic PETG**
2. Finish

**Verify the machine settings** (top-right printer dropdown -> the pencil/edit icon):

| Setting | Value |
| --- | --- |
| Bed size | 220 x 220 mm |
| Max height | 250 mm |
| Nozzle diameter | **0.4 mm** |
| Extruder type | **Direct Drive** <- this one matters most |

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# PART 2 - The settings that fix your problems

Switch OrcaSlicer to **Advanced** mode (top-right dropdown: Simple / Advanced / Expert -> choose **Advanced**), or you won't see half of these.

## 2.1 Quality tab

| Setting | Value | Why |
| --- | --- | --- |
| Layer height | **0.2 mm** | Good balance. 0.16 for the nicest surface, 0.28 if you're in a hurry. |
| First layer height | **0.25 mm** | Better bed adhesion |
| **Wall loops** | **5** | [!!] **THE FIX FOR YOUR BROKEN SCREW COLUMNS.** Default is 2. At 0.4mm each, 5 walls = 2mm of solid shell - on a dia 7.8mm boss that is nearly the entire column. |
| Top shell layers | **5** | Fixes the "exposed infill / rough top" you saw. Default 3 is too few. |
| Bottom shell layers | **4** | |
| Seam position | **Aligned** | Puts the layer-change blob in a consistent line instead of scattering it |

## 2.2 Strength tab

| Setting | Value | Why |
| --- | --- | --- |
| Sparse infill density | **25 %** | Plenty. With 5 walls, infill is not carrying the load. |
| Sparse infill pattern | **Gyroid** | Strong in all directions, no sharp direction changes to shake the printer |

## 2.3 Support tab - [!!] TURN IT ALL OFF

| Setting | Value |
| --- | --- |
| **Enable support** | **OFF (unchecked)** |

**Every overhang in these parts is already solved in the CAD:**

- Magnet pockets are **teardrop-shaped** so they self-support
- Pogo windows have **0.6mm printed-in bridges**
- The wire-exit hole has a **printed-in bridge** (added 2026-07-30 after your feedback - that

one genuinely was missing, you were right)

- Corner bosses have **gusset cones** that widen toward the floor

> [!] **These bridges are meant to be removed.** After printing, push them out with a small screwdriver or snip them with flush cutters. They are thin (0.6mm) and pop out easily. If you leave them in, the pogo pins and wires won't fit through.

**If you print with supports ON, the slicer fills the whole 60x60x50mm cavity with lattice that you cannot reach.** That is exactly what happened to you.

## 2.4 Speed tab

| Setting | Value | Why |
| --- | --- | --- |
| Outer wall | **30 mm/s** | Slower = better surface |
| Inner wall | 45 mm/s | |
| Sparse infill | 60 mm/s | |
| **Travel** | **150 mm/s** | Fast travel = less time to ooze = less stringing |
| First layer | **20 mm/s** | Adhesion |

## 2.5 Filament tab - [!!] THE STRINGING FIX

| Setting | Value | Why |
| --- | --- | --- |
| Nozzle temperature | **235 degC** (first layer 240) | If you still get strings, drop to **230**. Too hot is the #1 cause. |
| Bed temperature | **80 degC** | |

| Setting | Value | Why |
|---|---|---|
| **Retraction length** | **1.0 mm** | [!!] **Kobra Neo is DIRECT DRIVE.** Most guides say 5-6mm - that is for Bowden printers and will jam yours. |
| Retraction speed | **35 mm/s** | |
| Z-hop when retracting | **0.2 mm** | Stops the nozzle dragging through what it just printed |
| Wipe on retract | **ON** | |

### *[!!] The thing that actually causes stringing: wet filament*

**PETG absorbs moisture from the air, and Patiala is humid.** Wet filament strings no matter how perfect your settings are - the water boils out of the nozzle as steam.

**Dry it before printing:**

- Oven at **60-65 degC for 4-6 hours** (door slightly ajar). Do **not** exceed 70 degC or the spool warps.
- Or a food dehydrator at 65 degC
- Store it afterwards in a sealed box with silica gel

If the filament crackles or pops as it extrudes, it is wet. That is your problem, not settings.

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# PART 3 - Making JUST the screw columns solid

You asked exactly the right question: you want the screw columns denser without making the whole part 100 % infill. This is a **modifier**.

**Honest answer first:** with **Wall Loops = 5** the columns are already nearly solid, and that alone will probably fix the breakage. Try that before bothering with modifiers.

If you still want them fully solid:

1. Load `outer_box.stl`, **right-click** the model
2. **Add modifier** -> **Box**
3. A translucent box appears. In the right-hand **Object list**, select it.
4. Set its **size and position** so it encloses one corner boss:

- Size: `12 x 12 x 40` mm
- Position: `X = 26`, `Y = 21`, `Z = 20` (that is one boss; the four are at X +/-26, Y +/-21)

1. With the modifier still selected, in the parameter panel below, click **+ Add setting** ->

**Strength** -> **Sparse infill density** -> set it to **100 %**

1. **Repeat for the other three corners** (or select the modifier, Ctrl+C / Ctrl+V, and change

the position)

Anything inside that box prints at 100 % infill; everything else stays at 25 %.

> The same trick works for the mid-plate ledge or any other feature you want beefed up.

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# PART 4 - Orientation, part by part

[!!] `outer_box` **must be printed OPEN SIDE UP.** The rim sits on the bed, cavity facing the ceiling.

> A previous version of our own notes said to flip it rim-down. **That was wrong and I've corrected it.** Nothing can bridge a 60mm cavity, and flipping it hangs the four dia 7.8 x 37mm bosses tip-first in mid-air - which is very likely why yours snapped.

| Part | Orientation | Supports | Layer | Infill | Walls |
|---|---|---|---|---|---|
| **outer_box** | **Open side UP**, rim on the bed | [X] OFF | 0.2 | 25 % | **5** |
| **base_plate** | Flat, standoffs UP | [X] OFF | 0.2 | 30 % | 4 |
| **mid_plate** | Flat, collar UP | [X] OFF | 0.2 | 30 % | 4 |
| **top_plate** | Flat, dot side UP, skirt down | [X] OFF | 0.16 | 30 % | 4 |

| Part | Orientation | Supports | Layer | Infill | Walls |
|---|---|---|---|---|---|
| **esp32_pod_shell** | Open side UP | [X] OFF | 0.2 | 25 % | **5** |
| **esp32_pod_lid** | Flat, ridges UP | [X] OFF | 0.2 | 25 % | 4 |
| **pogo_end_cap** | Flat | [X] OFF | 0.16 | 30 % | 3 |

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# PART 5 - Producing the G-code

1. **Load** the STL: `File -> Import -> Import STL`, pick e.g. `cad/stl/outer_box.stl`
2. **Orient it** per Part 4. `Rotate` tool on the left toolbar; hold the mouse on a face and

use **Place on face** to drop a chosen face onto the bed.

1. **Check the settings** from Parts 2-3
2. **Slice** - the `Slice plate` button, bottom right
3. **Inspect the preview.** This is the step that would have saved your first print:

- Drag the **vertical layer slider** on the right up and down
- Look for **blue/teal support material** - if you see any, supports are still on. Turn them off.
- Check the bridges over the pogo slots and wire exit actually appear as thin flat spans
- Confirm the corner bosses look solid, not hollow

1. **Export G-code** -> save to an SD card as e.g. `outer_box.gcode`
2. Hand the SD card to the lab. They just load and print.

> **Print ONE part first** - `pogo_end_cap` or `mid_plate`. Small, fast, and it tells you whether your settings are right before you commit 3 hours to the box.

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# PART 6 - After the print

| Job | How |
|---|---|
| Remove the sacrificial bridges | Push out with a small flat screwdriver, or snip with flush cutters. Pogo windows (x2) and wire exit (x1). |
| Clean the screw holes | The M2.5 pilots print ~0.2-0.3mm undersize. Run a **2.5mm drill bit** through by hand - do not power-drill, you'll melt it. |
| Magnet pockets | Should be clear (teardrop shape). If tight, a **8mm** bit turned by hand. |
| Stringing wisps | Snip with flush cutters, or wave a heat gun / lighter **quickly** past them (do not hold it there) |
| Layer lines | 400 -> 800 grit wet sandpaper if you care cosmetically |

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# PART 7 - How to close the Brain Pod

The pod is two printed parts: **shell** (the open box) and **lid**.

```
                 +----------------------+
                 |   POD LID            |  <- 3 ridges on top (tactile ID)
                 |  o                o  |  <- 2 screw holes at X = +/-27.5
                 +--+----------------+--+
                    |                |      the lid's front/back SKIRT
+-------------------+----------------+---+  drops over the shell and
|  POD SHELL                             |  self-locates it
|    [ESP32 sits on header strips]       |
|  ?                                 ?   |  <- 2 screw BOSSES, also at X = +/-27.5
+----------------------------------------+
```

**Steps:**

1. ESP32 plugged into its header strips inside the shell, wires routed
2. Lower the lid straight down - the **skirt on the front and back edges** slides over the

shell walls and centres it. The left/right ends stay flush (those are the docking faces).

1. The two holes line up over the two bosses
2. Drive **2 x M2 x 8mm self-tapping screws** down through the lid into the bosses

**Screw length matters:** the lid is 4mm thick and the boss is 8mm deep. An **M2x8** leaves 4mm of thread biting into the boss. An M2x6 only gives 2mm and will strip out - do not use it.

**No glue.** The pod is meant to open so you can swap the ESP32.

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## Appendix A - Cura equivalents

| OrcaSlicer | Cura |
|---|---|
| Wall loops | **Wall Line Count** |
| Sparse infill density | **Infill Density** |
| Enable support | **Generate Support** |
| Retraction length | **Retraction Distance** |
| Top shell layers | **Top Layers** |
| Modifier box | **Per Model Settings** -> *Modify settings for overlaps* |
| Slice plate | **Slice** |

In Cura you must also set the cog icon -> **Setting Visibility -> Expert** or most of these stay hidden.

## Appendix B - Manual Kobra Neo profile (only if it's not built in)

Add Printer -> Custom / Add local printer:

| | |
|---|---|
| Build volume | 220 x 220 x 250 mm |
| Nozzle | 0.4 mm |
| Filament | 1.75 mm |
| Extruder | **Direct drive** |
| Bed shape | Rectangular, origin front-left |
| Firmware / G-code flavour | **Marlin** |
| Start G-code | leave the default Marlin one |

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## The five-line version

1. **Supports OFF** - every overhang is already handled in the CAD
2. **Wall loops = 5** - this is what fixes the broken screw columns
3. **Retraction 1.0mm** - Kobra Neo is direct drive, not Bowden
4. **Dry the PETG** - 60-65 degC for 4-6 h; wet filament strings regardless of settings
5. **outer_box prints OPEN SIDE UP** - never rim-down